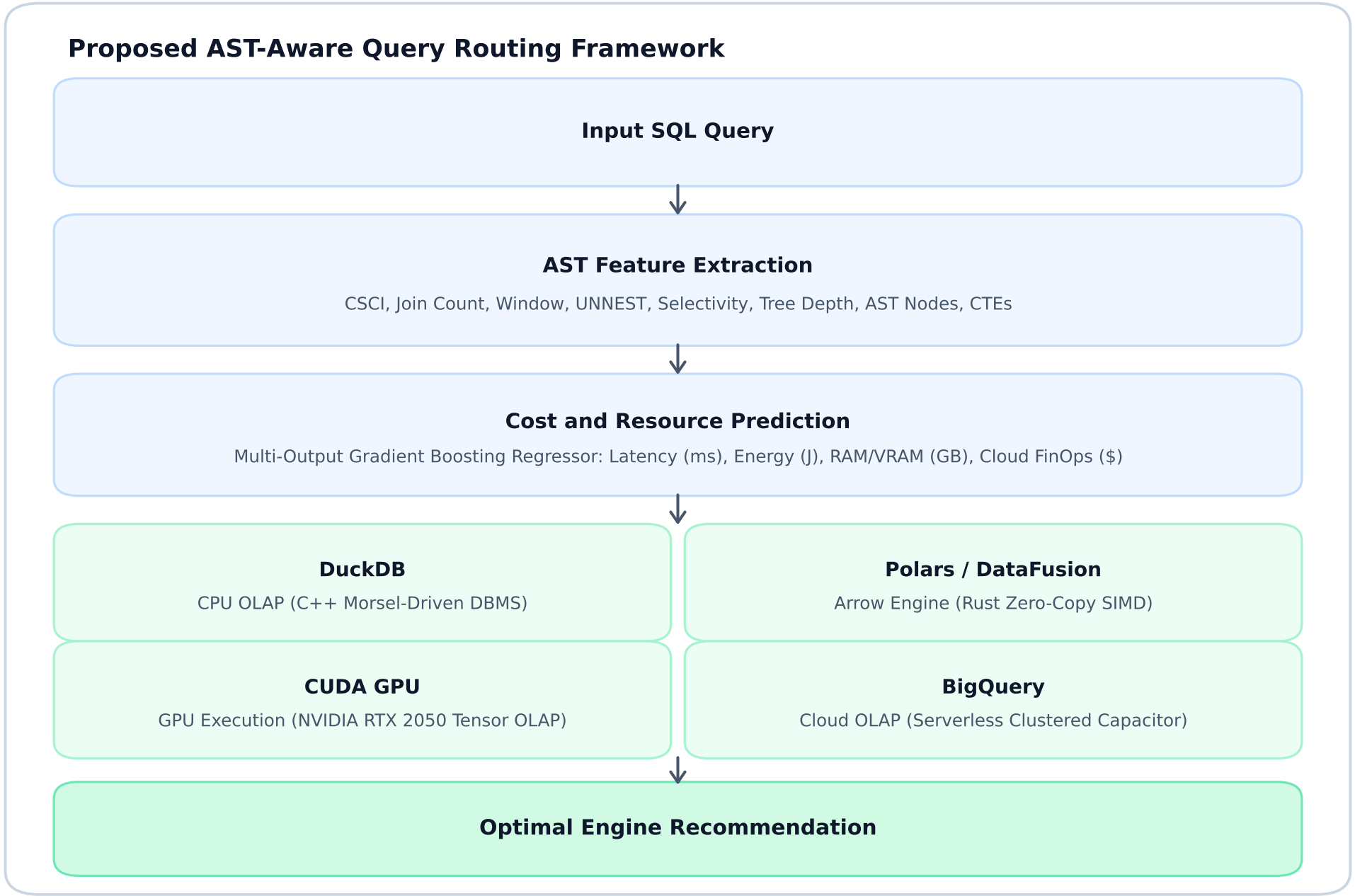
Proposed AST-Aware Query Routing Framework
Input SQL Query
AST Feature Extraction
CSCI, Join Count, Window, UNNEST, Selectivity, Tree Depth, AST Nodes, CTEs
Cost and Resource Prediction
Multi-Output Gradient Boosting Regressor: Latency (ms), Energy (J), RAM/VRAM (GB), Cloud FinOps ($)
DuckDB
CPU OLAP (C++ Morsel-Driven DBMS)
Polars / DataFusion
Arrow Engine (Rust Zero-Copy SIMD)
CUDA GPU
GPU Execution (NVIDIA RTX 2050 Tensor OLAP)
BigQuery
Cloud OLAP (Serverless Clustered Capacitor)
Optimal Engine Recommendation